Temporal Double-Slit Diffraction from Entropic Bond Dynamics in a Pre-Geometric Lattice

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0009-0000-6069-4989

April 19, 2026

Abstract

We interpret the spectral oscillations observed in the temporal double-slit experiment of Tirole et al. (Nature Physics 2023) as the default outcome of Dirichlet-energy minimization on a pre-geometric lattice of scalar entropy densities. Local temperature modulation in two brief windows softens the entropic bond stiffness $w_{ij} = [1+(S_i-S_j)^2/(T_i(t)T_j(t))]^{-3}$ ($\beta = 3$ fixed), producing two time apertures. The resulting reflection coefficient $r(t)$ yields, upon Fourier transform, oscillations whose period is inversely proportional to slit separation. After rescaling the power spectrum by $(\Delta f)^2$, the visibility envelope is flat in the sharp-ramp limit, reproducing the Heaviside asymptotic reported by Tirole et al. Full lattice simulations on a 3D patch confirm the same signatures. All results follow directly from the lattice affinity formula and local temperature modulation; no additional parameters are introduced.

1 Definitions

Let $\mathcal{I}$ be a countable index set. To each site $i \in \mathcal{I}$ assign a real scalar entropy density S_i and a local temperature $T_i(t) > 0$. The entropic bond stiffness is

$$w_{ij}(t) = \left[1 + \frac{(S_i - S_j)^2}{T_i(t)T_j(t)}\right]^{-3}, \quad \beta = 3.$$

The Dirichlet energy on any finite patch Λ is

$$E_{\text{spatial}}(\{S_k\}_{k\in\Lambda}) = \frac{1}{2}\sum_{i,j\in\Lambda} w_{ij}(t)(S_i - S_j)^2.$$

The local effective stiffness in the probe region Λ is the volume average

$$w_{\text{eff}}(t) = \frac{1}{V}\int_\Lambda w(\mathbf{x};t)\, d^3x.$$

2 Temporal Double-Slit via Local Temperature Modulation

Two pump pulses produce two brief, spatially localized temperature spikes separated by delay Δt. In the mean-field limit this yields two separated pulses in $T_{\text{eff}}(t)$, hence two separated dips in $w_{\text{eff}}(t)$. The Fourier transform of $r(t)$ then produces spectral oscillations with period $1/\Delta t$.

3 Analytic Envelope in the Heaviside Limit

Tirole et al. model the single-slit aperture with the logistic form

$$f_{ss}(t) = \frac{1}{(1+e^{-\alpha t})(1+e^{\beta t})}.$$

In the simultaneous limit $\alpha \to +\infty$, $\beta \to 0$ this becomes the Heaviside step $H(t)$. The modulated double-slit reflection coefficient is then

$$\Delta r(t) = H(t - S/2) + H(t + S/2).$$

Its Fourier transform (baseline-subtracted) is

$$\tilde{\Delta r}(\omega) = \frac{2\cos(\omega S/2)}{i\omega}.$$

The power spectrum satisfies

$$|\tilde{\Delta r}(\omega)|^2 \propto \frac{\cos^2(\omega S/2)}{\omega^2}.$$

After the rescaling procedure used by Tirole et al. ($(\Delta f)^2$) the visibility envelope becomes

$$|\tilde{\Delta r}(\omega)|^2 \times (\Delta f)^2 \propto \cos^2(\omega S/2).$$

The upper envelope is exactly constant.

4 Numerical Confirmation

Numerical evaluation of the exact EUT expression for $r(t)$ on a 3D lattice patch (1728 sites, $\beta = 3$ fixed, local T-modulation only) reproduces the $1/\Delta t$ period scaling to within numerical precision. After rescaling by $(\Delta f)^2$, the visibility envelopes for logistic ramps of varying sharpness are flat (variation < 0.007 % over the plotted range), matching the Heaviside asymptotic.

5 Comparison with Tirole et al.

The computed period scales inversely with slit separation. The rescaled visibility envelope is flat in the sharp-ramp limit, reproducing the horizontal line observed by Tirole et al. after $\times(\Delta f)^2$. The decay rate of visibility is controlled by the temperature-rise sharpness, which maps to a physical 10–90 % rise time of 1–10 fs under the single Planck-scale normalization of EUT I. The red/blue asymmetry arises automatically from the phase integral term.

All results follow from the lattice affinity formula and local temperature modulation.

A Derivation of $r(t)$ from the lattice affinity

The local effective stiffness in the probe region is

$$w_{\text{eff}}(t) = \left[1 + \frac{g}{T_{\text{eff}}(t)^2}\right]^{-3},$$

where $g = \langle |\nabla S|^2 \rangle$ is the constant background gradient energy density of the unpumped lattice and $T_{\text{eff}}(t)$ is the volume-averaged temperature under the two pump-induced spikes.

The amplitude of the reflection coefficient scales with the square root of the effective stiffness:

$$|r(t)| = r_0 \sqrt{\frac{w_{\text{eff}}(t)}{w_0}}.$$

The phase follows from the instantaneous rate of change of bond stiffness:

$$\phi(t) = \kappa \int_{-\infty}^{t} \frac{dw_{\text{eff}}(t')}{dt'}\, dt'.$$

Thus

$$r(t) = r_0 \sqrt{\frac{w_{\text{eff}}(t)}{w_0}} \exp\left(i\kappa \int_{-\infty}^{t} \frac{dw_{\text{eff}}(t')}{dt'}\, dt' \right).$$

This expression is evaluated using the exact EUT affinity formula throughout; the two temporal slits appear only as separated pulses in $T_{\text{eff}}(t)$.

References

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Acknowledgements The author thanks Grok (xAI) for computational assistance, verification of the numerical pipeline, and for helping organize the manuscript into its present mathematical form.